APPLIED MATHEMATICS
(CLASS 12)

DETAILED BREAKDOWN OF SYLLABUS

Ch. No.	Chapter Name	Topics
1	Numbers, Quantification and Numerical Application	5 Topics
2	Numerical Inequalities	5 Topics
3	Matrices	13 Topics
4	Determinants	8 Topics
5	Differentiation	4 Topics
6	Applications of Derivatives	9 Topics
7	Integrals	10 Topics
8	Differential Equations	5 Topics
9	Probability	7 Topics
10	Inferential Statistics	12 Topics
11	Time Based Data	5 Topics
12	Financial Mathematics	6 Topics
13	Linear Programming	4 Topics
Total	**13 Chapters**	**93 Topics**

Aman Kumar

Aniket
Date
Page 1

Applied Mathematics

Ch 1: Numers, Quantification and Numerical Application

T1: Modulo Arithmetic
- congruence modulo
- Equivalance class
- Properties
- Modulo with exponent

T2: Alligation and Mixture
- Formula of alligation
- Formula of Repeated dilution

T3: Boats and Streams

T4: Pipes and cisterns

T5: Races and Games.

Ch 2: Numerical inequalities
- meaning
- Properties
- In one variable
- In two variable
- Modulus function

Ch 3: Matrices
T1. meanig
T2. Types of matrices
T3. operations
T4. Multiplication of matrices

T5. 5 Properties of matrix
T6 multiplication
T7. Power of a square matrix
T8. Polynomial
T9. Transpose of a matrix
T10. Properties of Transpose
T11. Symmetric matrices
T12. Skew-symmetric matrices
T13. Sum of above both

ch4: Determinants

T1: meaning and value
T2: Minors and cofactors
T3: Expansion of Δ by any row or column
T4: 4 properties
T5: Adjoint and its properties
T6: Singular and inverse matrix
T7. Properties
T8: Matrix method or Martin's rule

ch5: Differentiation

T1: Revision of 11th
T2: Logarithmic functions
T3: Parametric forms
T4: Higher order derivatives

Ch6: Applications of Derivatives

T1. Derivative as a rate measure
T2. Increasing and decreasing function
- methods of Intervals
- conditions

T3: Maxima & Minima (values)
T4: Local maxima & Local minima
- critical points
- rules

T5: Local Maximum and Local minimum values
- cases
- Tests

T6. Marginal cost and Marginal revenue
T7. Fixed cost, variable cost, Average cost.
T8. Demand functions, Revenue & Profit functions and Breakeven analysis
T9. Marginal functions.
- MC
- MAC
- Relation b/w MC & AC

Ch 7: Integrals

T1: Indefinite integrals
- meaning and definition
- Anti derivative and primitive concept
- constant of integration
- Geometric interpretation of indefinite integrals

T2 Elementary standard integrals
- Power rule for integration

- Logarithmic integral
- Exponential integral
- Standard algebraic integral formulas

T3: Properties of idefinite Integrals
- Linearity properties of integrals
- Sum & difference rules
- Scalar multiplication rules
- Generalised properties for multiple functions

T4: Integration by substitution
- Concept of variable substitution
- Standard substitution rules
- Integrals of functions of linear forms
- special forms and selection of proper substitution.

T5: Integration by Partial fractions
- Proper and Improper rational functions
- Decomposition into partial fractions.
- Types of partial fractions

T6: Some more standard integrals
- Integrals involving square roots of quadratic expressions
- Standard formulae for irrational algebraic forms.

T7: Integration by parts
- Product rule for integration
- ILATE rule for choosing the first function.

- Special exponential form $\int e^x(f(x)+f'(x))\,dx$

T8: Definite Integrals
- Area function concept
- First fundamental theorem of integral calculus
- Second fundamental theorem of integral calculus
- Evaluation of definite integrals

T9: Application of integrals – Areas of Bounded Regions
- Area of the region bounded by a curve and coordinate axes.
- Area between two curves
- Symmetrical closed areas.

T10: Application of Integrals in Commerce and Economics
- Finding cost function from marginal cost
- Finding revenue function from marginal revenue.
- Law of demand and supply curves
- Equilibrium point
- Consumer surplus (CS)
- Producer surplus (PS)

Ch 8: Differential Equations

T1: Introduction and Basic concepts
- meaning and definition of differential eqn.
- Ordinary differential eqn.
- Distinction between independent and dependent variables.

T2: Order and Degree of a Differential eqn.
- Definition of the order of a differential equation.
- Definition of the degree of a differential equation.
- Polynomial conditions for determining degree.
- Examples of non-defined degrees.

T3: Solution of a Differential Equation
- Concept of explicit and implicit solutions.
- General solutions and arbitrary constants.
- Particular solutions.

T4: Formation of Differential Equations
- Eliminating arbitary constants from a given relation.
- Relationship between the number of constants and order of derivation.
- Conditions for a valid formed differential equation.

T5: Solving first order and first degree Differential equations.
- Overview of solvable forms
- Differential equations with Variables Separable technique.
- Integration of separated terms and general solution with constant.

Ch9: Probability

T1: Random variables
- meaning
- Domain and Range
- Deskreate and continuous

T2: Probability Distribution of Random Variables
- Concept
- Conditions
- Comulatide probabilities

T3: mean and variance of probability

T4: Binomial experiment and Binomial Distribution
- Bernoulli trials - Characteristics
- Formula
- Parameters n and p
- mean and variance

T5: Poisson Distribution
- Introduction to Poisson Distribution as alimiting form of binomial distribution.
- Parameter λ
- Recurrence formula
- mean and variance

T6: Continuous probability Distribution

T7: Normal Distribution
- Gaussian distribution and normal curve
- Standard normal distribution and standard Normal variable.

- Area under Standard normal curve and cumulative distribution function F(z).
- Properties of F(z)
- Area under any normal curve using z.

z-score formula $z = \frac{x - \mu}{\sigma}$

Ch 10: Inferential statistics

T1:
- Population and sample
- meaning and definition of population in statistics
- Types of population
- Concept and role of a sample

T2:
- Sampling method:
- Probability sampling = (simple random, systematic, stratified, and cluster sampling)
- Non-probability sampling (convenience, voluntary response, Judgement, and snow ball sampling)

T3:
- Representative and non-Representative samples.
- Characteristics of representative samples.
- Characteristics and Pitfalls of non representative sample

T4:
- Sampling Errors and Point Estimation
- Nature of sampling Errors
- Point estimation of population Parameters using sample statistics

T5: Unbiased and Biased Sampling
- Probability of unbiased Sampling via probability methods
- Concepts of biased Sampling via non-probability methods.

T6: Sampling Distribution and Standard Error
- Concept of Sampling distribution of statistics
- Expected value and Variance of Sample means
- meaning and calculation of Standard Error of mean. (SEM) $\sigma_{\bar{x}}$

T7: Central limit theorem (CLT)
- Standard significance of CLT for sample means ($n \geq 30$)
- Behavior of Sampling distribution for non-normal populations.

T8: Interval Estimation
- Concept of confidence interval, confidence level, and confidence coefficient
- Margin of error calculation when population Standard deviation (σ) is known.

T9: Hypothesis testing Basics
- Formulation of Null Hypothesis (H_0) and Alternative Hypothesis (H_a)
- Types of testing situations (Research hypothesis, Validity of claims, decision making).
- One-tailed and two-tailed tests
- Type-I error (α) and Type II error β

T10: Hypothesis test for population mean (σ known)

- Test statistic formula (Z-score)
- Critical value approach and rejection rules for lower tail, upper tail, and two tailed test.

T11: T-test & T-Distribution

- Overview of t-tests (one sample, two sample, and paired t-tests)
- Properties of student's t-distribution and degree of freedom (n-1)

T12: One sample t-test (σ unknown)

- t test statistics formula
- Critical value approach for one sample mean testing using t-tables.

Ch 11: Time based data

T1: Time series

- meaning and definition of time series
- Real life examples and application in statistics, weather forecasting, and economics.

T2: Components of Time series

- trend component (T)
- cyclical component (C)
- seasonal component (S)
- Irregular component (I)

T3: Models of Time series Analysis
- Additive model ($O = T + C + S + I$)
- multiplicative model ($O = T \times C \times S \times I$)

T4: Trend Analysis and method of moving Averages
- Linear and non-linear
- Concept of moving averages to reduce fluctuations.
- Odd-Period moving averages (3 yearly, 5 yearly)
- Even period moving averages and centered moving averages (4-yearly).

T5: Method of least squares
- Principle of least squares and line of best fit Condition ($\Sigma(y - y_i) = 0$ and $\Sigma(y - y_i)^2$ is least).
- Equation of trend line ($y = a + bx$)
- Normal equations for fitting linear trend
- Simplification when mid-point of time is taken as origin ($\Sigma x = 0$)
- Rules for selecting origin when n is odd ~~versus~~ versus when n is even.

Ch2. Financial mathematics.

T1: Perpetuity
T2: Sinking funds
T3: Bonds and Bond valuation
T4: E.M.I.
T5: CAGR
T6: Depreciation

Aniket
Date
Page 52

Ch 13: Linear programming

T1: Introduction & Terminology
- meaning of LPP, objective function, and decision variables.
- Constraints, non-negativity, and feasible or infeasible regions.

T2: Mathematical formulation
- Setting up decision variables and constraints.
- Formulating maximization and minimization objective functions.

T3: Graphical method
- Plotting constraints and identifying feasible region
- corner point method and optimal solutions

T4: Types of LPP problems.
- maximization and minimization problems
- unique, multiple, unbounded, and infeasible solutions.